

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

REAR DIFFERENTIAL CONTROL MODULE (RDCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Rear Differential Control Module (RDCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Rear Drive Axle and Differential](#) (205-02)

Rear Drive Axle/Differential, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0562-16	System Voltage Low - Circuit voltage below threshold	Rear differential control module power or ground circuit open circuit, high resistance	Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the rear differential control module power and ground circuits for open circuit, high resistance
P0563-17	System Voltage High - Circuit voltage above threshold	Battery/charging system fault	Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the relevant section of the workshop manual and test the battery and charging system
P0604-00	Internal Control Module Random Access Memory (RAM) Error - No sub type information	Rear differential control module internal failure	Using the manufacturer approved diagnostic system, re-configure the rear differential control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0605-00	Internal Control Module Read Only Memory (ROM) Error - No sub type information	Rear differential control module internal failure	Using the manufacturer approved diagnostic system, re-configure the rear differential control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0606-47	Control Module Processor - Watchdog/safety MicroController failure	Rear differential control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module

P0607-49	Control Module Performance - Internal electronic failure	▪ Rear differential control module power or ground circuit open circuit, high resistance ▪ Rear differential control module internal failure	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the rear differential control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module
P062F-00	Internal Control Module EEPROM Error - No sub type information	▪ Rear differential control module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0652-16	Sensor Reference Voltage B Circuit Low - Circuit voltage below threshold	NOTE: Circuit reference HALL_V+ ▪ Rear differential actuator position sensor supply circuit short circuit to ground ▪ Rear differential control module internal failure	▪ Refer to the electrical circuit diagrams and check the rear differential actuator position sensor supply circuit for short circuit to ground ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0653-17	Sensor Reference Voltage B Circuit High - Circuit voltage above threshold	NOTE: Circuit reference HALL_V+	▪ Refer to the electrical circuit diagrams and check the rear differential actuator position sensor supply circuit for short circuit to power ▪ Using the manufacturer approved diagnostic system,

		▪ Rear differential actuator position sensor supply circuit short circuit to power ▪ Rear differential control module internal failure	clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0657-13	Actuator Supply Voltage A Circuit / Open - Open circuit	NOTE: Circuit reference MOTOR+ / MOTOR- ▪ Rear differential actuator power or ground circuit open circuit, high resistance	NOTE: Ignore P0657-13 if P0605-00 is also set (perform the corrective action for P0605-00) ▪ Refer to the electrical circuit diagrams and check the rear differential actuator power and ground circuits for open circuit, high resistance
P0658-11	Actuator Supply Voltage A Circuit / Low - Circuit short to ground	NOTE: Circuit reference MOTOR+ / MOTOR- ▪ Rear differential actuator power or ground circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear differential actuator power and ground circuits for short circuit to ground
P0659-12	Actuator Supply Voltage A Circuit / High - Circuit short to battery	NOTE: Circuit reference MOTOR+ / MOTOR- ▪ Rear differential	▪ Refer to the electrical circuit diagrams and check the rear differential actuator power and ground circuits for short circuit to power

		actuator power or ground circuit short circuit to power	
P080A-54	Clutch Position Not Learned - Missing calibration	Rear differential control module internal failure	NOTE: The rear differential control module performs a calibration routine every 500km. If any rear differential control module faults are present this routine will not run and P080A will set. Investigate other rear differential control module related DTCs first. P080A will also set if the rear differential control module connector has been removed and reconnected Investigate other rear differential control module related DTCs first. Using the manufacturer approved diagnostic system, perform routine - Control Module Calibration. Clear the DTCs and retest. If the fault persists, install a new rear differential control module
P0810-62	Clutch Position Control Error - Signal compare failure	NOTE: Circuit reference MOTOR+ / MOTOR- / HALL_V+ / HALL_IN1 / HALL_IN2 / HALL_GND Rear differential	- Refer to the electrical circuit diagrams and check the rear differential actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear differential actuator position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Rear differential actuator position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Rear differential actuator mechanical failure	■ Using the manufacturer approved diagnostic system, perform routine - Control Module Calibration. Clear the DTCs and retest. If the fault persists, install a new rear differential actuator
P0CBC-11	Drive Motor "A" Temperature Sensor Circuit - Circuit short to ground	NOTE: Circuit reference TEMP1+ ■ Rear differential actuator temperature sensor circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the rear differential actuator temperature sensor circuit for short circuit to ground
P0CBC-15	Drive Motor "A" Temperature Sensor Circuit - Circuit short to battery or open	NOTE: Circuit reference TEMP1+ / TEMP1- ■ Rear differential actuator temperature sensor circuit short circuit to power, open circuit, high resistance	■ Refer to the electrical circuit diagrams and check the rear differential actuator temperature sensor circuit for short circuit to power, open circuit, high resistance
P186A-13	Differential Lock-up Actuator		■ Refer to the electrical circuit diagrams and check the rear

	Brake Control Circuit / Open - Circuit open	NOTE: Circuit reference MAG+ / MAG- Rear differential actuator brake circuit open circuit, high resistance	differential actuator brake circuit for open circuit, high resistance
P186B-11	Differential Lock-up Actuator Brake Control Circuit Low - Circuit short to ground	NOTE: Circuit reference MAG+ / MAG- Rear differential actuator brake circuit short circuit to ground	Refer to the electrical circuit diagrams and check the rear differential actuator brake circuit for short circuit to ground
P186C-12	Differential Lock-up Actuator Brake Control Circuit High - Circuit short to battery	NOTE: Circuit reference MAG+ / MAG- Rear differential actuator brake circuit short circuit to power	Refer to the electrical circuit diagrams and check the rear differential actuator brake circuit for short circuit to power
P186D-97	Clutch Actuator Stuck - Component or system operation obstructed or blocked	NOTE: Circuit reference MOTOR+ / MOTOR-	NOTE: If additional rear differential related DTCs are also set, perform the relevant corrective action(s) first

		▪ Rear differential actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear differential actuator mechanical failure	▪ Refer to the electrical circuit diagrams and check the rear differential actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance (>1.8 Ohms) ▪ Using the manufacturer approved diagnostic system, perform the guided diagnostic routine. Clear the DTCs and retest. If fault persists, install a new rear differential actuator
P1888-11	Differential Oil Temperature Sensor Circuit Failure - Circuit short to ground	NOTE: Circuit reference TEMP SIG 2+ ▪ Rear differential oil temperature sensor signal circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor signal circuit for short circuit to ground
P1888-15	Differential Oil Temperature Sensor Circuit Failure - Circuit short to battery or open	NOTE: Circuit reference TEMP SIG 2+ / TEMP SIG 2- ▪ Rear differential oil temperature sensor signal or earth circuit short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor signal and earth circuits for short circuit to power, open circuit, high resistance
P188A-98	Differential Oil Temperature Too High/Too Low - Component or system over temperature	▪ Insufficient rear differential oil ▪ Incorrect rear differential oil specification ▪ Rear differential oil	

		temperature sensor signal circuit short circuit to ground	NOTES: ▪ This DTC is set when the rear differential oil temperature exceeds 160°C ▪ If P0605-00 is also set, perform the relevant corrective action(s) first ▪ Refer to the relevant section of the workshop manual and check the rear differential oil quantity ▪ Refer to the relevant section of the workshop manual and check the rear differential oil specification ▪ Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor signal circuit for short circuit to ground
P18A5-12	Motion Position Sensor "A" Signal Circuit - Circuit short to battery	NOTE: Circuit reference HALL_IN1 / HALL_IN2 ▪ Rear differential actuator position sensor signal 1 circuit short circuit to power ▪ Rear differential actuator position sensor signal 2 circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the rear differential actuator position sensor signal 1 circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check the rear differential actuator position sensor signal 2 circuit for short circuit to power
P18A5-	Motion Position		▪ Refer to the electrical circuit

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P18A8-74	Differential Actuator Brake Control - Actuator slipping	▪ Rear differential actuator brake mechanical failure	▪ Using the manufacturer approved diagnostic system, perform the guided diagnostic routine. Clear the DTCs and retest. If fault persists, install a new rear differential actuator
P2787-4B	Clutch Temperature Too High - Over temperature	▪ Rear differential actuator active for prolonged duration due to driving style or terrain	NOTES: ▪ This DTC is set when the rear differential clutch pack calculated temperature exceeds 200°C ▪ If P0605-00 is also set, perform the relevant corrective action(s) first ▪ If additional rear differential temperature related DTCs are also set, perform the relevant corrective action(s) first

			Allow the rear differential to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication - Bus off	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Rear differential control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the rear differential control module with the latest level software
U1A14-49	CAN Initialization Failure - Internal electronic failure	Rear differential control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module

U2000-4B	Motor Temperature - Over temperature	▪ Rear differential actuator temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear differential control module internal failure	NOTES: ▪ This DTC is set when the rear differential actuator temperature exceeds 150°C ▪ If P0605-00 is also set, perform the relevant corrective action(s) first ▪ Refer to the electrical circuit diagrams and check the rear differential actuator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module
U2011-1C	Motor - Circuit voltage out of range	NOTE: Circuit reference MOTOR+ / MOTOR- ▪ Rear differential actuator positive circuit short circuit to actuator negative circuit ▪ Rear differential control module internal failure	▪ Refer to the electrical circuit diagrams and check the rear differential actuator positive circuit for short circuit to the actuator negative circuit ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module
U210A-62	Temperature Sensor - Signal compare failure	▪ Mismatch between rear differential oil temperature sensor signal and rear	▪ Allow the rear differential to cool (1 hour minimum). Using the manufacturer approved diagnostic system, check

		differential actuator temperature sensor signal - Insufficient rear differential oil - Incorrect rear differential oil specification - Rear differential oil temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear differential actuator temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	datalogger signals - Motor Temperature (0xD11C) - Rear Differential Sump Oil Temperature (0x1E8A). Difference should be <25°C - Refer to the relevant section of the workshop manual and check the rear differential oil quantity - Refer to the relevant section of the workshop manual and check the rear differential oil specification - Refer to the electrical circuit diagrams and check the rear differential oil temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the rear differential actuator temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U3000-4B	Control Module - Over temperature	- Rear differential control module insecurely installed - Rear differential control module internal failure	NOTE: This DTC is set when the rear differential control module internal temperature exceeds 105°C - Check the security and integrity of the rear differential control module fixings - Allow the rear differential control module to cool (1 hour minimum). Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear differential control module

